

Generalized Probabilistic Topological Field Theories: Notes

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INTRODUCTION

THE CATEGORY OF GPTS

Generalized probabilistic theories (GPTs) capture the minimal structure a physical theory needs to make probabilistic predictions [1–4]; classical probability and quantum theory are special cases. The theme of this section: that structure is linear algebra plus positivity. A system is a real vector space with a cone and a normalization functional, a transformation is a linear map preserving them, and composites are built on the tensor product. Every definition below is a **Vec** datum plus a positivity condition. We follow Müller [3] and Plávala [4]; for the categorical viewpoint see [5, 8].

Objects: systems

A *state* is an equivalence class of preparation procedures (Fig. 1): two preparations are identified when no measurement distinguishes their statistics [3]. Mixing makes the set of states convex, hence embeddable in a real vector space on which outcome probabilities are linear functionals; closure under limits and boundedness of probabilities make it compact; filtering (success probability $\lambda \leq 1$) closes it under nonnegative rescaling. The primitive carrier is thus the cone of *unnormalized* states; normalization survives as a single functional. Throughout, vector spaces are finite-dimensional, real, and spanned by the states they carry.

A *proper cone* $C \subseteq V$ is closed, convex, stable under nonnegative scaling, pointed ($C \cap (-C) = \{0\}$), and generating ($\text{span} C = V$): in order, closure of the state space, mixing and filtering, no vector physical together with its negative, no directions unrelated to physics.

Definition (System). A system is a triple $A = (V_A, V_A^+, u_A)$: a finite-dimensional real vector space V_A , a proper cone $V_A^+ \subseteq V_A$ of unnormalized states, and a unit $u_A \in V_A^*$ with $u_A(v) > 0$ for all $v \in V_A^+ \setminus \{0\}$.

The unit grades the cone by total weight (Fig. 2): $u_A(v)$ is the probability that the preparation v succeeds. Normalization is therefore implicit — the *subnormalized* states are $\{v \in V_A^+ : u_A(v) \leq 1\}$, the *normalized* ones the slice

$$\Omega_A := \{\omega \in V_A^+ : u_A(\omega) = 1\},$$

a compact convex base of the cone: every $v \in V_A^+ \setminus \{0\}$ is uniquely $u_A(v)\omega$ with $\omega \in \Omega_A$, strict positivity of u_A being exactly what lets every unnormalized state be normalized. *Pure* states are the extreme points of Ω_A . Alternatively, take (V_A, Ω_A) as primitive and recover cone and unit [3].

Effects. Measurement outcomes need no new data: outcome probabilities are linear in the state, so outcomes are elements of V_A^* and the probability rule is the evaluation pairing of **Vec**. The *effect cone* is the dual cone

$$(V_A^+)^* := \{e \in V_A^* : e(v) \geq 0 \ \forall v \in V_A^+\},$$

the *allowed effects* are $\mathcal{E}_A := [0, u_A]$, so that $e(\omega) \in [0, 1]$ on Ω_A , and a *measurement* is a finite family $(e_i) \subset \mathcal{E}_A$ with $\sum_i e_i = u_A$: outcome probabilities $e_i(\omega)$ are nonnegative and sum to one. By the bipolar theorem $((V_A^+)^*)^* = V_A^+$: states are exactly the vectors nonnegative against all effects. Presentations treating states and effects as co-equal primitives coupled by an abstract pairing reduce to this one once effects are required to separate states and conversely.

No-restriction. Setting $\mathcal{E}_A = [0, u_A]$ declares every admissible effect physical: the *no-restriction hypothesis* [5], our default. Dropping it [6] adds one datum, an effect cone $W_A^+ \subseteq (V_A^+)^*$ spanning V_A^* and containing u_A ; everything below goes through with $(V_A^+)^*$ replaced by W_A^+ .

Examples (Fig. 3).

- *Classical:* $V = \mathbb{R}^N$, $V^+ = \mathbb{R}_{\geq 0}^N$, $u(p) = \sum_i p_i$. Ω is the probability simplex, its N vertices the deterministic distributions; effects are the vectors in $[0, 1]^N$. The orthant is self-dual.
- *Quantum:* $V = H_N(\mathbb{C})$, the Hermitian matrices — a real vector space of dimension N^2 — with the PSD cone and $u = \text{tr}$. Ω is the set of density matrices and, via the Hilbert–Schmidt pairing, $\mathcal{E}_A$ the POVM elements $0 \leq E \leq 1$. The PSD cone is self-dual.

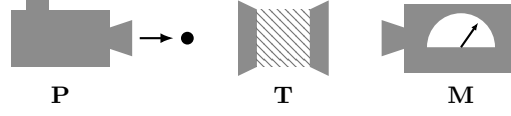


FIG. 1: The operational scenario (after [3]): a preparation device **P** emits a system, a transformation device **T** acts on it, a measurement device **M** registers an outcome a . A GPT specifies the devices and the probabilities $P(a|\mathbf{P},\mathbf{T},\mathbf{M})$.

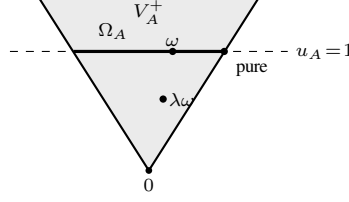


FIG. 2: A system: the cone V_A^+ of unnormalized states, graded by the unit. The slice $\Omega_A = u_A^{-1}(1) \cap V_A^+$ is the compact base of normalized states; its extreme points are pure, and $\lambda\omega$ is subnormalized.

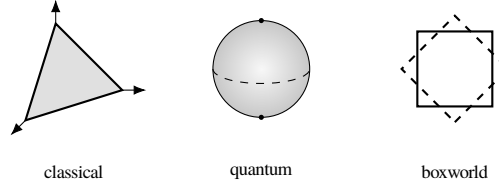


FIG. 3: Normalized-state slices. Classical: the simplex in the orthant. Quantum: the Bloch ball. Boxworld: the square, with the slice of its dual cone (dashed) — no linear map aligns the two.

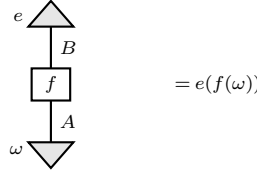


FIG. 4: Preparation, process and outcome as arrows, $\omega: \mathbf{1} \rightarrow A$, $f: A \rightarrow B$, $e: B \rightarrow \mathbf{1}$ — the categorical counterparts of the devices **P**, **T**, **M** of Fig. 1; their composite is the probability $e(f(\omega))$.

- *Boxworld (gbit)*: $V = \mathbb{R}^3$, V^+ the cone over a square [2]. The dual cone is the cone over the *rotated* square, and no linear map aligns the two: not self-dual [3, 10].

Self-duality of the state cone will decide the 1d GPTFT problem.

Morphisms: transformations

Mixing forces transformations to be linear [3]; the rest is preservation of the decorations.

Definition (Transformation). A morphism $f: A \rightarrow B$ is a linear map $f: V_A \rightarrow V_B$ with $f(V_A^+) \subseteq V_B^+$. It is allowed if $u_B \circ f \leq u_A$ on V_A^+ — equivalently, f sends subnormalized states to subnormalized states, the deficit being the failure probability — and a channel if equality holds. Composition and identities are those of $\mathbf{Vec}$.

The effect side takes care of itself: the transpose of a positive map is positive for the dual cones, since $(f^*e)(v) = e(f(v)) \geq 0$. With restricted effects, $f^*(W_B^+) \subseteq W_A^+$ is an independent condition and part of the definition.

$\text{Hom}(A, B)$ is closed under conic combinations but not subtraction: a cone inside the vector space $\text{Hom}_{\mathbf{Vec}}(V_A, V_B)$. GPT thus sits over $\mathbf{Vec}$ as a *wide, non-full* subcategory: all objects, only the positive maps.

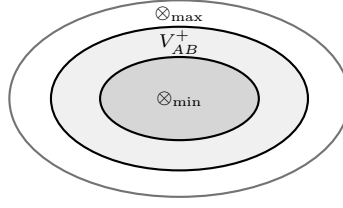


FIG. 5: The composite cone: any proper cone between the separable cone $\otimes_{\min}$ and $\otimes_{\max}$. For quantum systems, separable $\subsetneq$ PSD $\subsetneq$ witnesses.

The trivial system is $\mathbf{1} = (\mathbb{R}, \mathbb{R}_{\geq 0}, \text{id})$. A morphism $\mathbf{1} \rightarrow A$ is an element of V_A^+ , allowed iff subnormalized; a morphism $A \rightarrow \mathbf{1}$ is an element of $(V_A^+)^*$, allowed iff in $\mathcal{E}_A$. States and effects are ordinary arrows into and out of the unit (Fig. 4). In particular u_A is itself a morphism $A \rightarrow \mathbf{1}$, and channels are exactly the morphisms over $\mathbf{1}$: $u_B \circ f = u_A$.

Composites: the monoidal structure

Independent local preparations and measurements must combine into joint ones with factorizing statistics,

$$(e_A \otimes e_B)(\omega_A \otimes \omega_B) = e_A(\omega_A) e_B(\omega_B). \quad (1)$$

Mixing makes both products bilinear, so the joint space contains $V_A \otimes V_B$, with product states and effects the ordinary tensors [3]. Whether it is exactly $V_A \otimes V_B$ is a postulate:

Definition (Tomographic locality). *Joint states are determined by the joint statistics of local measurements [1, 2]; equivalently, $V_{AB} = V_A \otimes V_B$.*

We assume it by default. The local data bound the composite cone without fixing it: the *minimal tensor cone*

$$V_A^+ \otimes_{\min} V_B^+ := \text{cone}\{v \otimes v' : v \in V_A^+, v' \in V_B^+\}$$

is generated by the product states, while the *maximal tensor cone*

$$V_A^+ \otimes_{\max} V_B^+ := \{g : (e \otimes e')(g) \geq 0 \ \forall e \in (V_A^+)^*, e' \in (V_B^+)^*\}$$

contains everything consistent with product measurements. Both are proper, and duality exchanges them:

Lemma 1 (Min–max duality [4, 7]). *Under $(V_A \otimes V_B)^* \cong V_A^* \otimes V_B^*$, $(C_A \otimes_{\min} C_B)^* = C_A^* \otimes_{\max} C_B^*$ and $(C_A \otimes_{\max} C_B)^* = C_A^* \otimes_{\min} C_B^*$.*

Definition (Composite). *A composite of A and B is a system $AB = (V_A \otimes V_B, V_{AB}^+, u_A \otimes u_B)$ with*

$$V_A^+ \otimes_{\min} V_B^+ \subseteq V_{AB}^+ \subseteq V_A^+ \otimes_{\max} V_B^+. \quad (2)$$

Any intermediate proper cone is admissible (Fig. 5): a GPT includes a coherent (associative, symmetric) choice of composites, and any such choice makes $(\text{GPT}, \otimes, \mathbf{1})$ symmetric monoidal. The freedom in (2) is generic:

Theorem 1 (Entangleability of cones [7]). *$C_A \otimes_{\min} C_B = C_A \otimes_{\max} C_B$ iff C_A or C_B is simplicial (classical).*

Any two nonclassical systems therefore admit entangled states or effects. For quantum systems the three cones in (2) all differ: separable $\subsetneq$ PSD $\subsetneq$ entanglement witnesses. The quantum tensor product is an additional input — a choice strictly between the bounds [3].

No-signalling holds automatically: local effects sum to the local unit, so marginals on A do not depend on the measurement chosen on B ; the *reduced state* is $\omega_A := (\text{id}_A \otimes u_B)(\omega_{AB})$ [3].

Beyond tomographic locality. Without the postulate, the tensor product still embeds: $V_{AB} \cong (V_A \otimes V_B) \oplus C$, with a *global sector* C invisible to local tomography; $C = 0$ iff the composite is tomographically local. Real-vector-space quantum theory [9] and fermionic systems (parity superselection) have $C \neq 0$. The GPTFT constructions below do not require $C = 0$.

GPT versus Vec

Vec. An object is a finite-dimensional real vector space V ; a morphism is any linear map, so $\text{Hom}(V_A, V_B)$ is itself a vector space; the monoidal product is $\otimes$, with unit $\mathbb{R}$. The category is compact closed: every object has a dual V^* , the evaluation $\text{ev}(\varphi \otimes v) = \varphi(v)$ and coevaluation $\text{coev}(1) = \sum_a f_a \otimes f^a$ (dual bases) are canonical, and the snake identities hold for every V — there is never anything to check.

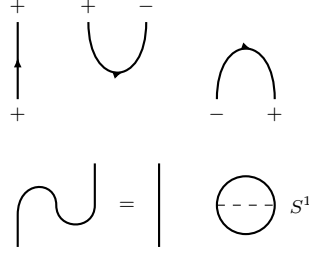


FIG. 6: Top: the generators of $\mathbf{Bord}_1$ — interval, cup, cap. Bottom: the snake relation (3), and the circle as a cup glued to a cap.

GPT. The same three layers, each carrying one positivity constraint (forgetting them is a functor back to $\mathbf{Vec}$):

$$\begin{aligned} V &\rightsquigarrow (V, V^+, u), \\ f &\rightsquigarrow f \text{ with } f(V_A^+) \subseteq V_B^+, \\ V_A \otimes V_B &\rightsquigarrow V_{AB}^+ \in [\otimes_{\min}, \otimes_{\max}], \quad u_A \otimes u_B. \end{aligned}$$

Each constraint changes the linear-algebraic character of its layer. The hom-set

$$\mathrm{Hom}_{\mathrm{GPT}}(A, B) = \{f \in \mathrm{Hom}_{\mathbf{Vec}}(V_A, V_B) : f(V_A^+) \subseteq V_B^+\}$$

is a cone, not a subspace: positivity of f does not give positivity of $-f$. The monoidal product is no longer a single functor but a family, the underlying space and unit fixed, the composite cone a choice in (2). Duals survive at the level of objects — swapping the state cone with the effect cone gives a dual system on V^* — but ev and coev are morphisms only if *positive*: the element $\sum_a f_a \otimes f^a$ must lie in the chosen composite cone, and in the dual of the cone chosen for the swapped pair. In $\mathbf{Vec}$ dualizability is automatic; in GPT it is a contingent property of the cones. Which GPTs pass is the subject of the next section.

1D TFTS

A topological field theory assigns algebraic data to manifolds, compatibly with gluing and disjoint union. In dimension one the whole theory is determined by a single object of the target, and its existence reduces to one categorical property of that object: *dualizability*. We fix the definitions for an arbitrary symmetric monoidal target $(\mathcal{C}, \otimes, \mathbf{1})$, then evaluate them in $\mathbf{Vec}$ and in GPT.

Definition (Bordism category). *Objects of $\mathbf{Bord}_1$ are closed oriented 0-manifolds: finite disjoint unions of the oriented points $+$ and $-$. Morphisms are compact oriented 1-bordisms up to diffeomorphism rel boundary, composed by gluing. Disjoint union $\sqcup$ is the monoidal product, the empty manifold $\emptyset$ its unit, exchange of factors the symmetry σ .*

As a symmetric monoidal category, $\mathbf{Bord}_1$ is generated by the interval $\mathrm{id}_+ : + \rightarrow +$, the cup $\cup : \emptyset \rightarrow + \sqcup -$, and the cap $\cap : - \sqcup + \rightarrow \emptyset$, subject to the *snake relations*

$$(\mathrm{id}_+ \sqcup \cap) \circ (\cup \sqcup \mathrm{id}_+) = \mathrm{id}_+, \quad (3)$$

$$(\cap \sqcup \mathrm{id}_-) \circ (\mathrm{id}_- \sqcup \cup) = \mathrm{id}_-, \quad (4)$$

an S-shaped strand pulls taut (Fig. 6). The only closed connected 1-manifold is the circle, a cup closed by a cap: $S^1 = \cap \circ \sigma \circ \cup \in \mathrm{End}(\emptyset)$.

Definition (TFT). A 1d TFT valued in $\mathcal{C}$ is a symmetric monoidal functor $Z : \mathbf{Bord}_1 \rightarrow \mathcal{C}$; in particular $Z(\emptyset) = \mathbf{1}$ and $Z(X \sqcup Y) = Z(X) \otimes Z(Y)$.

Definition (Dualizability). An object X of $\mathcal{C}$ is dualizable if there are an object $X^\vee$ and morphisms

$$\mathrm{ev} : X^\vee \otimes X \rightarrow \mathbf{1}, \quad \mathrm{coev} : \mathbf{1} \rightarrow X \otimes X^\vee$$

satisfying the snake identities (3)–(4) with $(\cup, \cap)$ replaced by $(\mathrm{coev}, \mathrm{ev})$. The duality data $(X^\vee, \mathrm{ev}, \mathrm{coev})$ are unique up to unique isomorphism. For dualizable X , the trace of $f \in \mathrm{End}(X)$ is $\mathrm{tr} f := \mathrm{ev} \circ \sigma \circ (f \otimes \mathrm{id}_{X^\vee}) \circ \mathrm{coev} \in \mathrm{End}(\mathbf{1})$, and $\dim X := \mathrm{tr} \mathrm{id}_X$.

Proposition 1 (Cobordism hypothesis, $d=1$ [11]). $Z \mapsto Z(+)$ is a bijection between isomorphism classes of 1d TFTs valued in $\mathcal{C}$ and of dualizable objects of $\mathcal{C}$, with $Z(-) = Z(+)^{\vee}$, $Z(\cup) = \mathrm{coev}$, $Z(\cap) = \mathrm{ev}$, and $Z(\bigsqcup_k S^1) = (\dim Z(+))^k$.

The categorical requirement on a 1d theory is thus exactly dualizability of its object at the point; all values on bordisms, and all closed invariants, follow.

1D TQFTs

In **Vec** every object is dualizable, with canonical data

$$V^\vee = V^*, \quad \text{ev}(\varphi \otimes v) = \varphi(v), \quad \text{coev}(1) = g := \sum_a f_a \otimes f_a$$

for dual bases $\{f_a\}, \{f^a\}$; g is id_V under $V \otimes V^* \cong \text{End}(V)$, hence basis-independent. The snakes reduce to the resolution of identity — tracing v through (3) gives $v \mapsto \sum_a f^a(v) f_a = v$ — and the circle computes the ordinary trace, $Z(S^1) = \sum_a f^a(f_a) = \dim_{\mathbb{R}} V$.

Proposition 2. *1d TQFTs valued in **Vec** are classified by the integer $n = \dim_{\mathbb{R}} Z(+)$; every closed invariant is a power n^k .*

Dualizability in **Vec** is a theorem: the data are canonical, the snakes are linear identities, nothing is left to check. This is exactly the freedom GPT withdraws.

1D GPTFTs

Definition (1D GPTFT). A 1d GPTFT is a symmetric monoidal functor $Z : \mathbf{Bord}_1 \rightarrow (\mathbf{GPT}, \otimes)$, the target carrying a coherent choice of composite cones (2); equivalently (Prop. 1), a dualizable object $X = Z(+)$ of GPT.

Positivity leaves no freedom in the candidate dual:

Proposition 3 (The dual is forced). *If $X = (V, V^+, u)$ is dualizable, then up to isomorphism*

$$X^\vee = (V^*, (V^+)^*, u^\vee),$$

with $u^\vee$ any interior point of V^+ , ev the canonical pairing, and $\text{coev}(1) = g$.

Proof. Forgetting cones sends duality data to duality data in **Vec**, which are canonical: identify the underlying space of $X^\vee$ with V^* , ev with the pairing, $\text{coev}(1)$ with g . Let W^+ be the state cone of $X^\vee$. Positivity of ev on $V_{X^\vee \otimes X}^+ \supseteq \otimes_{\min}$ gives $w(v) \geq 0$ for all $w \in W^+$, $v \in V^+$, i.e. $W^+ \subseteq (V^+)^*$. Membership $g \in V_{X \otimes X^\vee}^+ \subseteq \otimes_{\max}$ gives $(e \otimes x)(g) = e(x) \geq 0$ for all $e \in (V^+)^*$, $x \in (W^+)^*$, i.e. $(W^+)^* \subseteq V^+$, hence $(V^+)^* \subseteq W^+$. So $W^+ = (V^+)^*$. Units never enter the duality data; any interior point of V^+ (nonempty, V^+ being generating) is strictly positive on $(V^+)^*$ and makes $X^\vee$ a system. $\square$

Write $\psi := \text{coev}(1) = g$. Dualizability of X is the pair of conditions

$$\text{cup (state): } \psi \in V_{X \otimes X^\vee}^+, \tag{5}$$

$$\text{cap (effect): } \psi \in (V_{X^\vee \otimes X}^+)^*; \tag{6}$$

the snake identities, and $Z(S^1) = \dim_{\mathbb{R}} V$, then hold automatically, inherited from **Vec**.

Conditions (5)–(6) are a teleportation protocol (Fig. 7) [12]. Partition the snake as Bob = X_B , Alice = $X_A^\vee \otimes X_A$, and read the bordism along its time direction:

$t_0: \psi \in V_{X_B \otimes X_A^\vee}^+$	resource pre-shared,
$t_1: \psi \otimes v \in V_{X_B \otimes X_A^\vee \otimes X_A}^+$	input v at Alice,
$t_2: (\text{id}_{X_B} \otimes \text{ev})(\psi \otimes v) = \sum_a f^a(v) f_a$	Alice applies $\text{ev} = \psi$,
$t_3: = v \in V_{X_B}^+$	Bob holds v .

The transfer is exact, resource and measurement being one and the same ψ . Whether some admissible composite cone satisfies (5)–(6) — the cup preferring the cone large, the cap small — is the classification problem for 1d GPTFTs.

2D TFTs

In dimension two the objects are circles and the bordisms surfaces, and the classification again reduces the theory to one algebraic structure on $Z(S^1)$: a commutative Frobenius algebra internal to the target [13–15]. We record the **Vec** case; the GPT case is taken up afterwards.

Definition (Bordism category). *Objects of $\mathbf{Bord}_2$ are closed oriented 1-manifolds: finite disjoint unions $\bigsqcup_k S^1$. Morphisms are compact oriented surfaces up to diffeomorphism rel boundary, composed by gluing; $\square$ is the monoidal product, $\emptyset$ its unit, σ the exchange.*

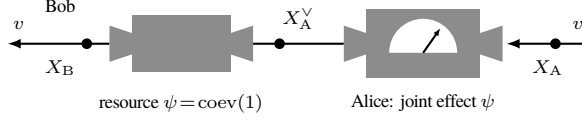


FIG. 7: The snake as exact teleportation, in the device style of Fig. 1. A source pre-shares the resource ψ between Bob (X_B) and Alice ($X_A^\vee$); Alice applies the joint effect ψ to the input v and her share; Bob's system exits in the state v .

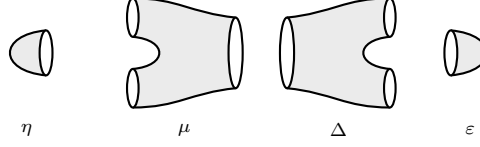


FIG. 8: Generators of $\mathbf{Bord}_2$: disk η , pants μ , reversed pants Δ , reversed disk ε (time flows left to right).

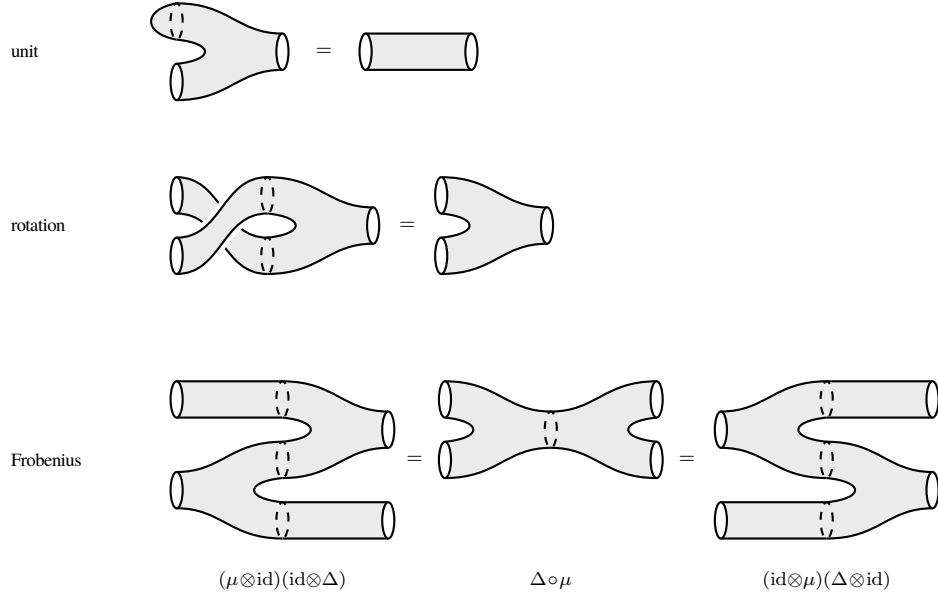


FIG. 9: Relations of $\mathbf{Bord}_2$ as diffeomorphisms; dashed circles are glued boundaries. *Unit*: capping a leg of the pants gives the cylinder. *Rotation*: the half-turn of the pants exchanges its legs, $\mu \circ \sigma = \mu$ (commutativity). *Frobenius*: the three pants decompositions of the two-in-two-out surface, $(\mu \otimes \text{id})(\text{id} \otimes \Delta) = \Delta \circ \mu = (\text{id} \otimes \mu)(\Delta \otimes \text{id})$, Eq. (7). Associativity and the co-versions are analogous.

Every compact oriented surface decomposes into disks, cylinders and pairs of pants. As a symmetric monoidal category $\mathbf{Bord}_2$ is generated by the cylinder id_{S^1} together with (Fig. 8)

$$\eta: \emptyset \rightarrow S^1, \quad \mu: S^1 \sqcup S^1 \rightarrow S^1, \quad \Delta: S^1 \rightarrow S^1 \sqcup S^1, \quad \varepsilon: S^1 \rightarrow \emptyset,$$

the disk, the pair of pants, the reversed pants, and the reversed disk. The relations are exactly the diffeomorphisms of glued surfaces (Fig. 9): μ is associative and commutative with unit η , Δ is coassociative and cocommutative with counit ε , and the cuttings of the two-in-two-out surface give the *Frobenius relation*

$$(\mu \otimes \text{id}) \circ (\text{id} \otimes \Delta) = \Delta \circ \mu = (\text{id} \otimes \mu) \circ (\Delta \otimes \text{id}). \quad (7)$$

2D TQFTs

Definition (Commutative Frobenius algebra). A commutative Frobenius algebra in $\mathbf{Vec}$ is a finite-dimensional commutative unital algebra $(A, \mu, 1_A)$ together with a counit $\varepsilon \in A^*$ whose induced pairing

$$\beta(a, b) := \varepsilon(\mu(a \otimes b))$$

is nondegenerate. The coproduct $\Delta: A \rightarrow A \otimes A$ is then the unique map dual to μ under β , and (7) holds.

Theorem 2 ([13, 14]). $Z \mapsto Z(S^1)$ is a bijection between isomorphism classes of 2d TQFTs valued in **Vec** and of commutative Frobenius algebras, with $Z(\eta), Z(\mu), Z(\Delta), Z(\varepsilon)$ the algebra data of the same name. The closed invariants are

$$Z(\Sigma_g) = \varepsilon(h^g(1_A)), \quad h := \mu \circ \Delta,$$

the handle operator; in particular $Z(S^2) = \varepsilon(1_A)$ and $Z(T^2) = \dim_{\mathbb{R}} A$.

Frobenius structure contains duality. Set $\gamma := (\Delta \circ \eta)(1) \in A \otimes A$ and $\beta = \varepsilon \circ \mu$. The snake identities for the pair (γ, β) follow from (7) and the (co)unit laws, so A is self-dual: $A \cong A^*$, with duality data internal to the theory. A 2d theory therefore contains the 1d problem for $X = Z(S^1)$ with $X \cong X^\vee$ — in GPT, all the positivity conditions (5)–(6) on ψ embed into the 2d problem, before any condition on μ and Δ is imposed.

2D GPTFTs

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